MILITARY SPECIFICATION

SEXTANT, BUBBLE AQU-1/A

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy

1 SCOPE

1.1 Scope - The equipment covered by this specification shall be a type of Bubble Sextant, AQU-1/A, for use in military aircraft.

1.2 Classification - The Bubble Sextant shall be of one type as specified in 6.4.

2 APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.5, of the issue in effect on the date of invitation for bids, form a part of this specification.

3 REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts and materials are such that their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specifications.

(3) When a nonstandard part is used, the meeting of general material requirements such as nonflammable, fungus resistance, nontoxic, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended must be approved first by the Bureau of Aeronautics.

FSC 6605

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3 1 1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows

(1) Mechanical parts shall be submitted as nonstandard parts only when the material requirements of Specification MIL-E-5400 are not met

(2) Nonstandard parts of an electrical nature as defined in Specification MIL-E-5400 must be approved before the preproduction model of the equipment is submitted for tests The request for approval shall be forwarded to the Bureau of Aeronautics before the part is purchased or fabricated In addition to or in lieu of the Nonstandard Part Data Sheet the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts

3 2 Design and Construction

3 2 1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein

3 2 2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 19 pounds

3 2 3 Operating Life

3 2 3 1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer

3 2 3 2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer

3 2 4 Cabling and Connection

3 2 4 1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400

3 2 4 2 Interconnecting Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088 The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment

3 2 4 3 Input Power Connections - Input power connections shall be in accordance with MS 25245

3 2 5 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307

3 2 6 Interchangeability - The equipment shall meet the interchangeability requirements as specified in Specification MIL-E-5400 Unless otherwise specified interchangeability shall be measured against a model supplied by the Bureau of Aeronautics

3 2 7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400 Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance

3 2 8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1 0 V AC and/or 27 5 ± 0 5 V DC

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3.2.9 Service Conditions - The equipment shall operate under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment except as specified herein.

3.2.10 DC Power 28-Volt System

Operating voltage limits	25 to 29 volts
Reduced performance voltage limits	20 to 25 and 29 to 31 volts
Current	1/4 Amps

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both the standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Sextant - hereinafter the term "sextant" shall be construed to mean the sextant proper, viewing system, control panel, connecting cable, and watch clip. The sextant shall be a bubble-type sextant in a sealed housing with an astrodome projecting above the skin of the aircraft. A mounting flange shall be provided which is integral with the sextant housing. Provisions shall be made for 360° rotation of the sighting prism, azimuth scale, and cursor. The presentation shall be such that the azimuth scale, lubber's line, cursor, altitude counter image, and averager period indicator, shall be visible in the eyepiece as shown in Figure 1.

3.3.2.1 Body - The body shall be constructed of either nonferrous low-density metal or synthetic material, uniform in texture with a smooth surface. Synthetic material shall be of the thermal setting type, composed of a suitable filler and phenol condensation product binder. The body shall be finished in durable dull black.

3.3.2.2 Bubble

3.3.2.2.1 Bubble Cell Replacement - The sextant shall be so designed and constructed that replacement of the bubble cell, together with the operation necessary to bring the sextant into complete operative adjustment, may be readily accomplished. The bubble cell shall be accurately constructed in order to provide interchangeability.

3.3.2.2.2 Bubble Size - The bubble size shall be such that its angular diameter in the field of view shall be 1°30' ± 20%.

3.3.2.3 Sextant Illumination - Artificial illumination of the water azimuth scale, lubber's line, cursor, bubble image, altitude counter and averager period indicator shall be provided through suitable arrangement of lights operating on approximately 28 volts, both A.C. and D.C., and so designed that the bulbs are easily replaced. The illumination of the presentation shall be such that the images are clearly defined with minimum glare in the field, and shall permit adequate dimming of the illumination for observation of second magnitude stars. The illumination shall be such that the presentation may be adequately illuminated for any combination of outside bright daylight and inside dim aircraft illumination levels.

3.3.2.3.1 Bubble Liquid - The liquid used in the bubble level shall conform to Specification MIL-F-5562. Other liquids may be used if specifically approved by the Bureau of Aeronautics.

3.3.2.4 Optical Requirements - The optical requirements shall be as follows:

(1) All lenses, prisms, and mirrors shall be of first quality, made from first-class stock and shall be accurately ground, polished, and centered. The optical system shall be well corrected for chromatic and spherical aberration. Lenses shall be mounted free from strain.

(2) The exit pupils for the bubble and for the star field shall be concentric within 4 millimeters.

(3) The eye distance for the star field and for the bubble field shall not differ by more than 12 millimeters.

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(4) Minute Indicator - The third drum shall contain numerals for each 5 minutes of altitude from 0 through 55. The height of these numerals shall be 3/32 inch. Graduation marks for each minute of altitude shall also appear on the third drum. The graduation marks for each minute of altitude shall be approximately 3/64 inch in length, with each consecutive 5-minute graduation mark approximately 5/64 inch in length, and both marks shall not exceed 0.005 inch in width.

(5) An index line engraved of 1/4 inch length and 0.01 inch width shall be vertically centered on the counter cover glass over the minute indicator, the cover glass in turn centered with respect to the counter drums. The index line shall be so situated to afford rapid alignment of the degree and minute values for accurate altitude reading.

3.3.2.6 Flexible Cable Drives - Flexible cable drives shall be secured with suitable brackets as required.

3.3.2.7 Filters and Shade Glasses - Suitable filters shall be provided and shall have a visible light transmission through a range of approximately 0.001% to 85.0%. Introduction of the shade glass into the line of sight shall not produce secondary images or deviation in excess of 30 seconds of arc along the line of sight or 3 minutes in a plane perpendicular to the line of sight.

3.3.2.8 Connecting Cable - The sextant shall be provided with a suitable connecting cable with a connector plug in accordance with Specification MIL-C-5015.

3.3.2.9 Watch Clips - A watch clip permanently fastened to the sextant, suitable for securely holding the watch, which shall meet the requirements of MS 28076, shall be so positioned on the sextant proper as to allow the watch face to be read with the eye in line with the eyepiece position. The watch shall be illuminated to permit adequate readability.

3.3.2.10 Variable Time Averager (VTA) - The variable time averager shall be designed to continuously average sextant altitude settings, and shall give the average altitude and half the elapsed time when used over any time interval from 30 seconds to 2 minutes. The averager shall be so constructed that no parts will work loose in service and shall be so built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage installation, and service, without failure. Backlash shall be reduced to a minimum.

(1) Dust Protection - The averager shall be enclosed in a dustproof housing.

(2) Winding Knob - The winding lever shall be located conveniently for winding the mechanism. One complete turn or less shall be sufficient to set the altitude averaging indicators to their respective zero positions. Approximately 5 seconds after release of the winding lever, the averager shall be ready for operation. The method of returning indicator to zero position shall not involve forced slippage of any part of the averager.

(3) Averager Zero Adjustment - After the mechanism has been rewound and the averager made ready for its intended usage, it shall be possible to adjust accurately the zero position of the averager by means of an adjustment screw accessible either from the outside of the case, or by removing the cover of the averager without disassembly of the mechanism.

(4) half-Time Dial - The diameter of the half-time dial shall be a minimum of 3/4 of an inch. It shall be graduated into 60 equal divisions, each division to represent 2 seconds of time. Every fifth graduation shall be a major graduation, identified by numerals. The numerals shall read from 0 to 55. The dial shall indicate elapsed time to within an accuracy of $\pm$ 4 seconds, or $\pm$ 2 seconds if read on the half-time dial. The timing mechanism shall drive the half-time dial to indicate elapsed time over a period of 2 minutes with a maximum allowable error of 2 seconds per minute.

(5) Averaging Indicator - An indicator shall be provided which, when returned to its zero position by operation of the sextant control knob upon completion of an observation and prior to rewinding, shall determine the average altitude, as read on the sextant altitude counter, to within 2 minutes of arc after a 2-minute observation without ambiguity throughout the total sextant altitude range.

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(4) The sextant presentation shall be fixed focus with the exit pupil 9 inches from the viewing lens. A rubber cone 8 inches in length shall be provided to eliminate stray light and facilitate use of the sextant. The cone shall be constructed such that it may be easily secured and detached from the sextant.

(5) The image of the celestial body and the bubble shall be brought into coincidence by reflection. Double images in the field of view shall be considered cause for rejection.

(6) Magnification of the optical system shall be unity.

(7) The sextant shall have a real field of not less than 14.5°, definition not being critical beyond 5° real field from the sighting axis. All internal optics except the etched reticle surface, and bubble chamber surfaces, shall be treated with a high-efficiency nonreflecting abrasion resistant coating.

(8) The angle of sight through the sextant proper shall be horizontal but the angle of sight through the viewing system shall conform to the particular aircraft requirements.

(9) The exit pupil shall be not less than 3/4 inch in diameter.

(10) Any loose foreign particles in the bubble cell or any evidence of stickiness or erratic behavior in the bubble cell during any test specified herein shall be considered cause for rejection.

(11) Every effort shall be made to keep foreign particles in the field of view to a minimum.

(12) An azimuth scale shall be provided such that its image is visible in the field of view. The graduations shall consist of one marker every 2° with each ten degree marker numerically identified. The azimuth scale shall be rotatable through 360° in the field of view.

(13) A reference line (lubber's line) shall extend into the field of view such that alignment of the reference line with the ship's axis will be facilitated. The overall accuracy of the true heading indication shall be 0.5°.

(14) The optical system of the sextant shall contain a cursor to facilitate collimation and the reading of true heading. The image of the cursor shall be visible in the exist pupil and rotatable through 360°. A lever shall be provided to permit rotation of the cursor relative to the azimuth scale or the azimuth scale and cursor relative to the lubber's line.

(15) The lubber's line, azimuth scale, heading cursor, altitude counter image, and averager indicator light shall have a common focal plane.

3.3.2.5 Altitude Range - The sextant shall be capable of measuring altitudes from 1° below the horizontal (zero) to 90° altitude. The altitude shall be indicated on a drum-type counter with one complete revolution of the operating knob indicating approximately 5° change in altitude.

(1) Altitude Counter - An altitude counter mechanism shall be incorporated.

(2) Counter Range - The counter shall indicate altitudes throughout the effective range of not less than -1° to +90°. Rugged stops shall be provided to prevent damage to the sextant, or to prevent disengagement of the operating parts when the operating knob is rotated in either direction beyond the operating range of the sextant. The design of the stops shall be such that forcible engagement of the stops impose no pressure between applicable operating parts of the sextant mechanism other than the normal pressure existing when the stops are disengaged.

(3) Degree Indicators - The counter shall consist of three adjacent drums. The first two drums shall contain numerals from 0 through 9, with a height of approximately 5/32 inch. The numerals and markings on the drums shall be of durable dull black on a durable white background. The numerals on the first two drums shall indicate altitude in degree.

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(6) Dial Marking—The markings of the half-time dial and the altitude averaging indicators shall have a background of durable dull black finish. Numerals and reference markings shall be painted white. When a revolving disk-type indication is used, a white rectangle approximately 1/8 by 1/4 of an inch shall be used in addition to the durable dull black for the background of the reference markers of the averaging indicator. The index mark on the white rectangle shall be a dull black.

(7) Operating Button or Lever - A single start-stop button or lever shall start and stop the operation of the averager. The button shall be so located that when an observer uses the sextant having the averager attached, he may conveniently operate the button with the thumb or one of the first two fingers of the hand not being used to operate the altitude control knob of the sextant.

(8) Period of Operation - When the averager is fully wound and ready for operation, the mechanism shall start and continue functioning when the start-stop lever is moved through not more than 3/4 full displacement. The lever may then be released without affecting the operation of the mechanism. The averager shall continue in operation through a period of 2 minutes ± 4 seconds and after operating for the maximum period, the averager shall stop automatically. After the averager has been operating for a period in excess of 30 seconds, depressing and releasing the button at any time before the total period has elapsed shall stop the operation.

A light shall be turned on when the start-stop lever is depressed to stop operation, or when the cycle of operation of the averaging device is complete. Within a period of 2 seconds after the end of the averager cycle, the averager shall actuate the light. The light shall be visible in the field of the azimuth scale, bubble and cursor.

(9) Range - The averager shall be capable of averaging a series of observations extending over the full sextant altitude range without limit stops.

(10) Windows - Suitable plastic windows shall cover both the half-time dial and the averaging indicators. An index line 0.02 inch wide shall be vertically centered on the window with respect to the zero position of the averaging indicators and half-time dial. That portion of the line extending over the averaging disks shall be durable dull black, while that portion of the line extending over the half-time dial shall be durable dull white.

(11) Desiccators - A desiccator shall be provided to prevent fogging of the astrodome and optical components. A suitable indicator shall also be provided to indicate the presence of moisture.

3.3.3 Pressure Seal - The mounting flange of the sextant shall be such that a suitable pressure seal will be effected when the sextant is installed in the aircraft. Any gaskets, if necessary, shall be furnished with the sextant.

3.3.4 Calibration Card - A calibration chart shall be attached to the housing of each sextant with the following information:

(1) Sextant index error at three points, 0, 45, and 90°

(2) The date of collimation of the sextant.

3.3.5 Cover Plate - A suitable cover plate shall be furnished under a separate part number to effect a pressure seal in the event a sextant is not installed in the aircraft.

3.3.6 Electrical Connectors - The electrical connectors shall be in accordance with Specification MIL-C-5015, except where otherwise noted.

3.4 Service Instructions - Unless otherwise specified, one (1) copy of operation and installation instructions shall be included with each sextant. Prior to publication, these instructions shall be approved by the Bureau of Aeronautics.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements.

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on two equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the Bureau of Aeronautics, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400.

4.2.3 Scope of Tests - Preproduction tests shall include all tests of 4.4.2 and any other tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Fabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract, however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected and satisfactorily pass the following tests conducted as specified in 4.4.

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(1) Examination of Product (see 4.4.2.1)

(2) Scale Error (see 4.4.2.2)

(3) Bubble Cell Curvature (see 4.4.2.3)

(4) Shade Glass (see 4.4.2.4)

(5) Fixed Setting (see 4.4.2.5)

(6) Slippage (see 4.4.2.6)

(7) Continuously Changing Settings (assembled sextant) (see 4.4.2.7)

(8) Variable Time Averager (VTA) - zero adjustment (see 4.4.2.8)

(9) Variable Time Averager (VTA) - zero setting (see 4.4.2.9)

(10) Variable Time Averager (VTA) - backlash in averaging mechanism (see 4.4.2.10)

(11) Maximum Operation Period (see 4.4.2.11)

(12) Dielectric Strength (see 4.4.2.12)

(13) Azimuth Mechanism Performance (see 4.4.2.13)

4.3.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the tests of 4.3.1. The following quantity of equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests conducted as specified in 4.4.

(1) Vibration Error Test

(2) Sextant and Bubble Cell Temperature

(3) Nonfogging

(4) Illumination

(5) Counter

4.3.3 Special Tests - Special tests shall be conducted as determined by the Bureau of Aeronautics on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4 Test Methods

4.4.1 Test Conditions - Tests shall be conducted under the standard conditions of 3.2.8 unless otherwise specified herein.

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4.4.1.1 Attitude - Unless otherwise specified, the sextant shall be tested in normal operating position.

4.4.1.2 Rate of Altitude Control Knob Rotation - Unless otherwise specified, the rate of rotation of the drive mechanism for operating the sextant knob shall in no case exceed 10° of altitude reading per second.

4.4.1.3 Test Measurements - All measurements shall be made with instruments whose accuracy has been verified. If tests are conducted at the contractor's plant the accuracy of the instruments and test equipment shall be verified periodically by the contractor to the satisfaction of the Bureau of Aeronautics.

4.4.1.4 Test Equipment - When a special chamber is required to accomplish the tests, the equipment shall be mounted in the chamber and electrical or mechanical connections completed where operation is required during the test period. Access covers and inspection plates shall be in place except when such covers would normally be removed for service use. When operation is not required, external connections shall be plugged to simulate actual service conditions.

4.4.1.5 Pressure - A pressure chamber shall be used which is capable of withstanding a pressure differential of 20 psi continuously without loss of pressure. Provisions shall be made for mounting the sextant in the chamber, with the sextant in the normal operating position. A suitable air flowmeter shall be used to measure the flow of air to the test chamber.

4.4.1.6 Averager - Unless otherwise specified, the averager shall be tested while attached to the sextant.

4.4.2 Tests -

4.4.2.1 Examination of Product Test - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.4.2.2 Scale Error Test - Each sample sextant shall be tested for scale error at five points: -1, 0, +1, 45, and 90 degrees of altitude with the bubble centered on the cross hairs, or at any other points selected at the discretion of the testing activity with both increasing and decreasing readings. The altitude error at any point shall not exceed 3 minutes of arc. The sextant shall be bench-tested to determine true heading error. True heading shall be checked each 90° from 0° to 360° with azimuth set in to 0.25°, and with the bubble in the center of the field. Total error in true heading shall not exceed 0.5 degree.

4.4.2.3 Bubble Cell Curvature Test - Each sample sextant shall be tested for corrections of bubble cell curvature. The error due to incorrect curvature of the bubble cell shall not exceed 2 minutes of arc at any point along the line of sight up to 2/3 of the distance from the center to the edge of the bubble field.

4.4.2.4 Shade Glass Test - Each sample sextant shall be tested to determine the error caused by the shade glasses. The use of any shade glass shall not introduce secondary images or deviation in excess of 30 seconds of arc along the line of sight or 3 minutes in a plane perpendicular to the line of sight.

4.4.2.5 Fixed Setting Test - The averager shall be operated for any desired period of time between 1 and 2 minutes on any altitude setting with the knob held stationary. Movement of the primary indicating element shall be not more than 0.01 inch. This test shall be performed three times using different operating periods and drum settings, arbitrarily chosen.

4.4.2.6 Slippage - The averager shall be operated for the maximum possible time interval and the altitude setting noted after bringing the indices into coincidence. Without rewinding the mechanism, the knob shall be rapidly oscillated 5° from its setting. After a minimum 5 cycles, the indicator shall be brought into coincidence. The altitude setting then obtained from the averager shall not differ from that previously obtained by more than 2 minutes of arc. To eliminate the effect of backlash from this test, both altitude readings shall be obtained as the primary element of the averaging indicator is brought into alignment by rotation of the knob in the same direction as that of the start of the cycles. This test may be combined with other tests if desired.

4.4.2.7 Continuously Changing Settings (Assembled Sextant) - The averager shall be operated for any desired period of time during which the altitude settings change continuously and at a constant rate, through a range of not more than 2.5 degrees. The average altitude computed by the averager shall agree with the average of the initial and final altitude settings to within the following:

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At the end of 30 seconds, ± 5 minutes of arc;

At the end of 1 minute, ±3 minutes of arc; and

At the end of 2 minutes, ±2 minutes of arc.

This test shall be performed five times using different operating periods and altitude settings arbitrarily chosen. At least one test shall include initial and final drum settings differing by approximately 2.5 degrees.

4.4.2.8 Variable Time Averager (VTA)-Zero Adjustment - After the instrument has been rewound, and before the averaging mechanism is put into operation, oscillation of the knob through 10° range of reading shall not move the element of the averaging indicator more than 0.01 inch for a series of 5 or more cycles.

4.4.2.9 Variable Time Averager (VTA)-Zero Setting - After the averaging device has operated for a complete cycle of operations, it shall be rewound. When rewound, the indicator shall return to within ± 0.01 inch of its zero index mark.

4.4.2.10 Variable Time Averager (VTA)-Backlash in Averaging Mechanism - After the instrument has run for the maximum time interval and before it is rewound, backlash in the averaging mechanism shall be determined by bringing the indicating elements and the index marker into coincidence first from one direction then from the opposite direction and noting the resulting sextant altitude readings. These readings shall not differ by more than 1 minute. To eliminate the effect of slippage all operations of this test shall be done slowly.

4.4.2.11 Maximum Operating Period - The averager shall be operated three times for the maximum time interval. The length of time of each such period shall be 2 minutes ±4 seconds. This test may be combined with other tests if desired.

4.4.2.12 Dielectric Strength - With the lamps removed and bubble disconnected, the sextant shall be subjected to a high potential test. When 500 volts RMS, 60 cycle A-C power is applied for 1 minute between "A" and "P" of the receptacles, there shall be no breakdown of insulation.

4.4.2.13 Azimuth Mechanism Performance - The azimuth dial shall be rotated for at least 1 minute in each direction at speed not to be in excess of 20° of azimuth indication per second. Care shall be taken to limit starting accelerations to the maximum obtainable by hand operation. No slippage of the azimuth mechanism shall be permitted.

4.4.2.14 Illumination - The sextant shall be tested to determine compliance with the requirements specified in 3.3.2.3 and 3.3.2.7.

4.4.2.15 Counter - The sextant shall be tested to determine compliance with the requirements specified in 3.3.2.5 through 3.3.2.5(2).

4.4.2.16 Filter Test - Conformance to 3.3.2.7 is required.

4.4.2.17 Environmental Test - Prior to starting the environmental tests a performance record shall be obtained as specified in Specification MIL-E-5272. During and after all environmental test operation shall not indicate sticking or erratic action. At the discretion of the testing activity, individual tests may be performed, after each environmental test, immediately after removing the sextant from the specified chamber. The following environmental tests shall be conducted:

4.4.2.17.1 High Temperature Test - This test shall be in accordance with Procedure I of Specification MIL-E-5272 with the following exception. After the chamber has been returned to room temperature and the sextant has saturated at room temperature for a minimum period of 4 hours, it shall meet the individual test requirements specified herein. There shall be no damage which would adversely affect subsequent operation.

4.4.2.17.2 Low Temperature Test - This test shall be in accordance with Procedure II of Specification MIL-E-5272 with the following exception. The sextant shall then be subjected to the individual tests specified herein while at the -54°C temperature. There shall be no damage which would adversely affect subsequent operation.

4.4 4.4.2.17.3 High Altitude Test - This test shall be conducted in accordance with Procedure II of Specification MIL-E-5272 for a continuous period of 4 hours. In addition, while at the specified pressure the instruments shall be operated and subjected to the individual tests specified herein. There shall be no failure as a result of this test.

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4.4.2.17.4 Nonfogging - The sextant shall be placed in a test chamber which is capable of being sealed, and in which the temperature and relative humidity have been raised to 22°C and 95 - 100% respectively. After 3 hours, the sextant shall be removed from the test chamber and placed in another chamber in which the temperature is -40°C. The pressure shall then be reduced to 9 inches Hg (approx. 30,000 feet altitude). After 1 hour in the cold chamber, the sextant shall be removed and so placed between heating elements that the metal case of the sextant receives the direct heat radiation. As the temperature of the sextant is raised, no fogging shall occur on the optical or glass surfaces.

4.4.2.17.5 Rain Test - The test shall be conducted in accordance with Specification MIL-E-5272. The sextant shall be mounted such that only the astro dome is exposed to the rain. The portion of the sextant which extends below the astrodome shall be protected from the rain. In mounting the sextant, external connections should be made to simulate installation conditions as near as possible. Duration of the test period shall be 8 hours, at the completion of which the sextant shall be examined and no evidence of water penetration or damage shall be visible. At the end of the test with the sextant installed, the mounting seal shall be examined and there shall be no evidence of water penetration.

4.4.2.17.6 Humidity Test - This test shall be in accordance with Procedure I of Specification MIL-E-5272.

4.4.2.17.7 Fungus Resistance Test - This test shall be in accordance with Specification MIL-E-5272.

4.4.2.17.8 Salt Spray Test - This test shall be in accordance with Specification MIL-E-5272. After removal of the salt encrustations, the sextant shall be subjected to the individual tests specified herein.

4.4.2.17.9 Vibration Tests - Vibration tests shall consist of the following two tests:

(1) Vibration Failure Test - This test shall be conducted as described herein: With the sextant mounted in its normal position, vibratory motions made on the sextant while installed in the mount shall be transmitted by the mount to the sextant when the mount is vibrated. Upon completion of the test, the sextant shall be subjected to and satisfactorily pass the individual and sampling tests. No looseness or damage to the mechanism shall result from the vibration. Resonant frequencies shall be determined by varying the frequency of applied vibration slowly through the frequency range of 10 to 500 cps at any vibratory accelerations not exceeding those shown in Figure 2. This procedure shall be followed successively for vibration applied along three mutually perpendicular axes of the test specimen. The test specimens shall be vibrated at the indicated resonant conditions for 60 minutes with the applied amplitude or vibratory acceleration specified in Figure 2. These periods of vibration shall be accomplished in sequence for each of 3 mutually perpendicular axes of vibration. When more than one resonant frequency is encountered with vibration applied along any one axis, the test period may be accomplished at the most severe resonance, or the period may be divided among the resonant frequencies, whichever is considered most likely to produce failure. When resonant frequencies are not apparent within the specified frequency range, the specimens shall be vibrated for a period of 2 hours at a frequency of 30 cps and an applied amplitude of 0.03 inch.

(2) Vibration Error Test - This test shall be in accordance with Procedure V of Specification MIL-E-5272 except that the frequency of vibration shall be varied uniformly from 500 to 3,000 cpm. Creepage of the azimuth scale, shall not exceed 0.5° during any 2-minute period of vibration.

4.4.2.17.10 Equipment Crash Safety - The sextant shall be tested for crash safety in accordance with Specification MIL-E-5272, using a dummy load inside of the main sextant housing. A dummy load shall also be used in place of the (VTA) variable time averager.

4.4.2.17.11 Sextant and Bubble Cell Temperature - The sextant shall be subjected to a temperature of -54°C for a period of 4 hours. The operating controls and moving parts of the sextant shall function properly while at this temperature. The sextant shall then be subjected to a temperature of +70 C for a period of 4 hours. No damage or leakage shall occur to the bubble cell as a result of

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this test. The operating controls and moving parts of the sextant shall function while at this temperature. The bubble shall be completely sealed. Bubble diameter at 20°C shall be approximately 1°30' and shall not vary by more than 20% in the temperature range -54°C to +70°C.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-I-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Bureau of Aeronautics.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's, and EMB's shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A - The sextant shall be packaged in accordance with Specification MIL-P-7936, Level A. Preservation shall be Method II D of Specification MIL-P-116 utilizing a reuseable container conforming to either MIL-C-6054, MIL-C-6055, or MIL-D-26943.

5.1.2 Level C - The sextant shall be packed for shipment in accordance with the standard commercial practice of this type instrument.

5.2 Packing

5.2.1 Level A & B - The sextant shall be packed in accordance with MIL-P-7936, Level A for overseas destinations and Level B for domestic destinations.

5.2.2 Level C - The sextant packaged in accordance with 5.1.1 or 5.1.2 shall be overpacked to conform with standard commercial practice.

5.3 Marking— All marking shall conform to the requirements of MIL-STD-129.

6. NOTES

6.1 Use - The sextant covered by this specification is intended for use in aircraft to measure the angular altitude of a celestial body by reference to a bubble artificial horizon and to measure true heading of aircraft.

6.2 Preproduction Tests - It is expected that the contract or purchase order will specify that two sextants, as applicable, will be required as preproduction samples and that these preproduction samples will be subjected to the preproduction tests to determine compliance with the requirements of this specification. The invitation for bids and the contract should specify the point of inspection for these tests. It is to be understood that items supplied under contract will be identical in every respect to the sample tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the samples will constitute cause for rejection.

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6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request shall be accompanied by complete supporting information.

6.4 Items of Equipment - The equipment covered by this specification shall consist of the following items:

One (1) sextant proper

One (1) viewing system

One (1) control panel

Cables, electrical and mechanical connecting, as necessary

One (1) watch clip

One (1) cover plate

6.5 Documents - Applicable documents are listed as follows:

Specifications

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors; Electrical, "AN" type
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing Environmental, Aircraft Electronic Equipment
MIL-F-5562	Fluid; Bubble Sextant
MIL-C-6054	Container: Steel Shipping
MIL-C-6055	Drums; Metal, with Removable Head, Reuseable Interior Shipping
MIL-P-7936	Parts and Equipment, Aeronautical Preparation for Delivery
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-D-26943	Drums; Metal, Reuseable Exterior Shipping

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

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MS-25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single-Phase AC, or 115/200 Volt 3-Phase AC, or combinations
MS-28076	Watch, Pocket, Navigation, Master

PUBLICATIONS

Bureau of Aeronautics

SAR-300	Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins, Preparation of

6.5.1 Availability of Documents:

(1) When requesting applicable documents give both the title and number. All requests shall be made via the cognizance Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, (CDS), Philadelphia 11, Pennsylvania.

6.5.2 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over all referenced documents.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Deviations from this specification, or from subsidiary documents where applicable, may be made only upon written approval.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 Definitions -

6.6.1 "Knob" - The term "knob", when used, has reference to the sextant altitude control knob or the equivalent rotating altitude control device on the test fixture.

6.6.2 "Altitude Setting" - The term "altitude setting," when used, is assumed to mean the provision made on either the sextant or test fixture for reading the angular altitude set in by the altitude control knob.

6.6.3 "Amplitude" - Whenever the word "Amplitude" is specified, it will mean the extent of motion measured from one extreme to the opposite extreme.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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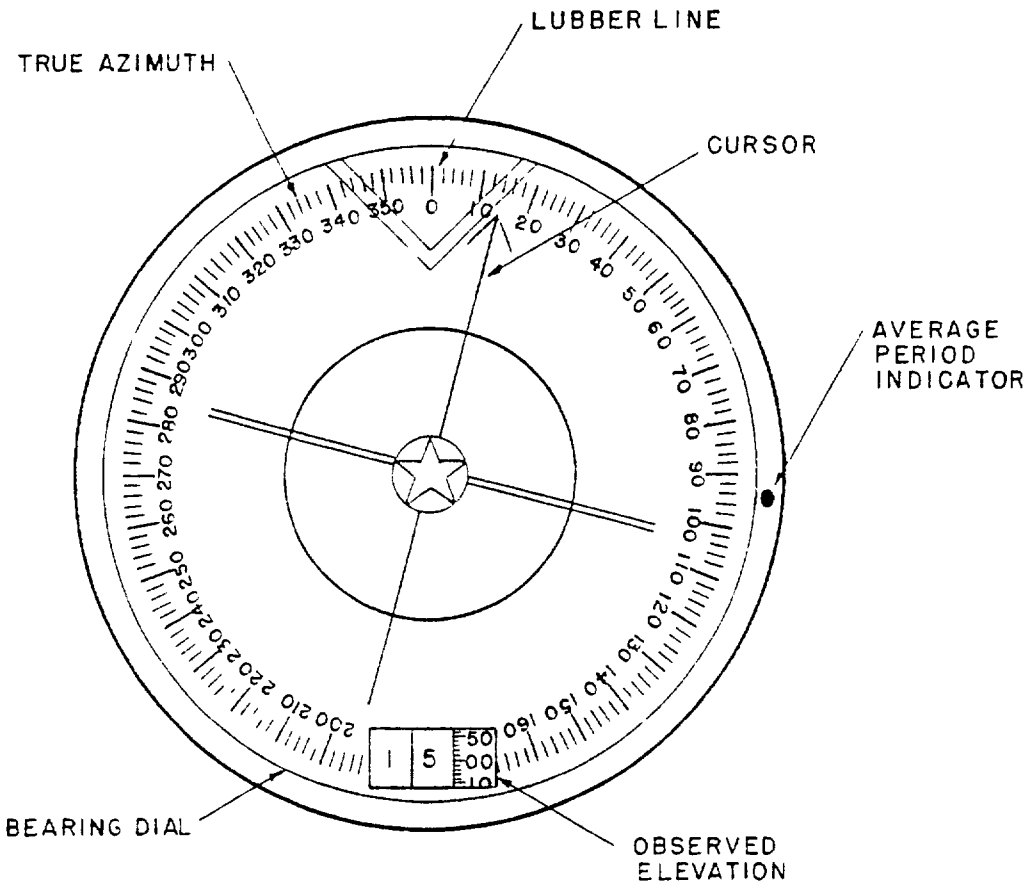


FIGURE 1 PRESENTATION

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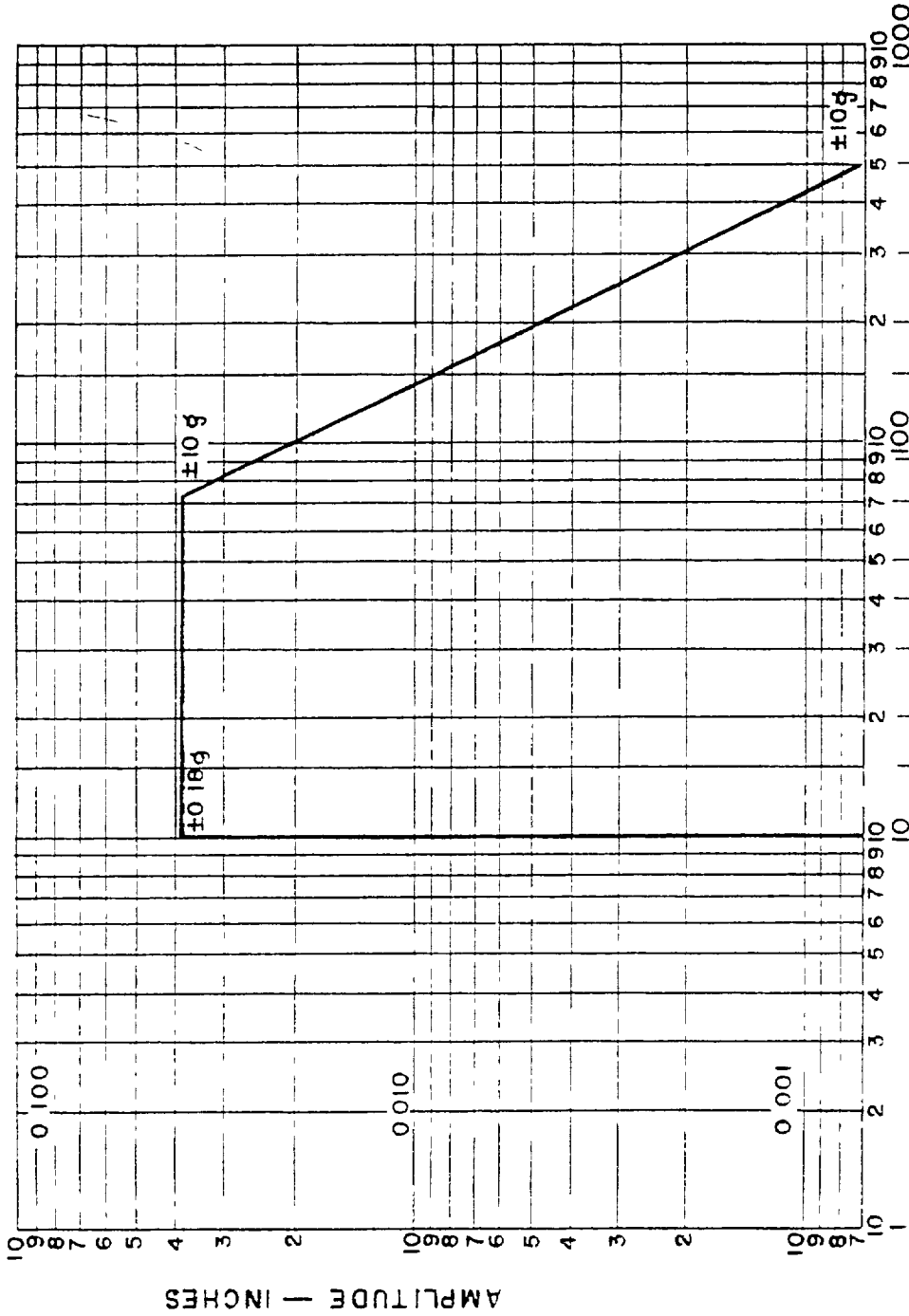


FIGURE 2 RANGE CURVE FOR VIBRATION TESTS

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